Checkpoint A - question paper

Attempt guide

Checkpoint A is a 60-minute, 30-mark cumulative diagnostic covering selected required material from Weeks 1–4. All questions are compulsory. It does not assess `M5`, optional dynamics or any excluded topic.

For this controlled first-cohort pilot, your score is diagnostic only. It is not used for admission, progression, ranking, funding or grading. Use it to identify revision priorities rather than treating the result as a pass/fail or ranking decision.

Attempt sequence

1. Open the question paper and formula sheet.
2. Complete the paper in 60 minutes without notes or solutions.
3. Use the full solutions to review the attempt and identify revision priorities.
4. Record the first cause of each lost mark and schedule a blank-page reattempt after two or three days.

Candidate paper

Time allowed: 60 minutes
Total marks: 30
Questions: All questions and all parts are compulsory.

Instructions

- Complete the paper without notes or solutions.
- A basic non-programmable scientific calculator is permitted, but no question requires one.
- Software, programmable calculators, computer algebra and statistical packages are not permitted.
- Show the reasoning and intermediate work needed to support each answer. An unsupported answer may not receive full marks.
- Give exact answers unless the question asks for an approximation.
- All vectors are column vectors. Use the course notation throughout.

Question 1 — reasoning, algebra and one-variable calculus (9 marks; 17 minutes)

Part A — `CPA-Q01-PA` (4 marks; 7 minutes) For $n \in \{1, 2, ...\}$ and $r \in \mathbb{R} \setminus \{1\}$, a student makes the claim

$$\sum_{j=0}^{n} r^j = \frac{1 - r^n}{1 - r}.$$

Write the claim using the quantifiers $\forall$, then write its logical negation using $\exists$. Disprove the claim with one admissible counterexample and state the correct finite-sum identity.

Part B — `CPA-Q01-PB` (5 marks; 10 minutes) Let

$$f(x) = -x^3 + 3x^2 + 9x, \qquad x \in [-2, 4].$$

Find all stationary points in the interval. Evaluate f at every stationary point and endpoint needed for a global comparison, then determine the global maximum of f on $[-2, 4]$, including where it is attained.

Question 2 — linear systems and multivariable differentiation (8 marks; 17 minutes)

Part A — `CPA-Q02-PA` (4 marks; 8 minutes) Use Gaussian elimination on the augmented matrix to solve

$$\begin{aligned} x + y &= 3, \\ 2x - y + z &= 4, \\ y + z &= 2. \end{aligned}$$

State the rank of the coefficient matrix and classify the system as having a unique solution, no solution or multiple solutions.

Show a traceable sequence of row operations. Substitution-only work does not earn the Gaussian-elimination method mark, although it may earn any separately evidenced solution, rank and classification marks.

Part B — `CPA-Q02-PB` (4 marks; 9 minutes) Define $F : \mathbb{R}^2 \to \mathbb{R}$ and $\mathbf{G} : \mathbb{R}^2 \to \mathbb{R}^2$ by

$$F(u, v) = u^2 + uv, \qquad \mathbf{G}(x, y) = \begin{bmatrix} x + y \\ x - y \end{bmatrix},$$

and let $h := F \circ \mathbf{G}$. Use the multivariable chain rule to calculate the column gradient $\nabla h(1, 0)$. State the dimensions of the derivative objects that you multiply, and verify the first component of your answer by differentiating the explicit composition directly.

Question 3 — probability rules and Bayes' rule (7 marks; 13 minutes)

Part A — `CPA-Q03-PA` (3 marks; 5 minutes) For events A and B, suppose

$$\mathbb{P}(A) = \frac{3}{5}, \qquad \mathbb{P}(B) = \frac{1}{2}, \qquad \mathbb{P}(A \cap B) = \frac{1}{4}.$$

Calculate the probability of each of the following events:

1. $A \cup B$;
2. exactly one of A and B; and
3. neither A nor B.

Part B — `CPA-Q03-PB` (4 marks; 8 minutes) A submitted application is high-priority with probability $1/4$. Let H denote the event that it is high-priority and S the event that an initial screen is positive. Suppose

$$\mathbb{P}(S \mid H) = \frac{4}{5}, \qquad \mathbb{P}(S \mid H^c) = \frac{1}{5}.$$

Calculate $\mathbb{P}(S)$ and $\mathbb{P}(H \mid S)$. Show the branches used in the total-probability calculation.

Question 4 — random variables and moments (6 marks; 13 minutes)

Part A — `CPA-Q04-PA` (3 marks; 5 minutes) Let X have probability mass function

$$p_X(x) = c(x + 1), \qquad x \in \{0, 1, 2\},$$

and $p_X(x) = 0$ otherwise. Find c, then calculate $\mathbb{P}(X \geq 1)$ and $F_X(1)$, where F_X is the cumulative distribution function of X.

Part B — `CPA-Q04-PB` (3 marks; 8 minutes) Random variables X and Y satisfy

$$\mathbb{E}[X] = 1, \quad \mathbb{E}[Y] = 2, \quad \mathrm{Var}(X) = 2, \quad \mathrm{Var}(Y) = 3, \quad \mathrm{Cov}(X, Y) = \frac{1}{2}.$$

For $Z = 2X - Y$, calculate $\mathbb{E}[Z]$ and $\mathrm{Var}(Z)$. Explain how the positive covariance affects $\mathrm{Var}(Z)$ relative to a calculation that incorrectly treats X and Y as independent.

Formula sheet

General conventions

- Give exact answers unless a question asks for an approximation.
- All vectors are column vectors.
- Use $\mathbb{P}(A \mid B)$ for conditional probability, $p_X(x)$ for a probability mass function and $F_X(x) = \mathbb{P}(X \leq x)$ for a cumulative distribution function.
- Use $\mathbb{E}[X]$, $\mathrm{Var}(X)$ and $\mathrm{Cov}(X, Y)$ for expectation, variance and covariance.

Formula provision

No numerical tables or critical values are required for this paper. No additional formulae are supplied because the finite-series, calculus, linear-system, multivariable-chain-rule and probability identities used in the questions are part of the assessed material.